Problem Set 2

Quantitative Investing (FINC-B8420), Fall 2026

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Due: Monday, October 19, 2026, 5pm

Instructions and Guidelines

- This problem set is due **Monday, October 19, 2026** by 5pm. Come to class (on 10/28) prepared to explain the decisions you made in constructing your estimations/forecasts.
- Turn in one problem set per group. Please submit to Canvas.
- You will again need to pull data from CRSP and Compustat using the WRDS interface. Please refer to Neel's writeup on data sources (in Files⇒ProblemSets) if you have questions about finding the data.
- The writeup has a page limit of 5 pages, not including tables and figures. You can include an appendix if you wish, but the writeup should be self-contained. That is, we should be able to figure out what you did and why you did it without referencing the appendix.
- In addition to the writeup, please submit your code (or a link to your group's git repository).
- Again, a reminder that this is a brand new problem set. If anything is unclear please reach out to Neel or Pono for clarification.

Use the database, characteristics, and factor returns you constructed in Homework 1, adding daily returns and other information as needed. Work within the July 1963 – December 2025 sample, allowing for the return history needed to initialize estimates and the availability of additional data.

Question 1 — Predicting Market Betas

A stock's market beta measures the sensitivity of its return to the return on the market. Let $r_{i,\tau}$ denote the return on stock i on day τ and let $r_{m,\tau}$ denote the market return.[1] Using excess returns, we have

$$\widetilde{r}_{i,\tau} = r_{i,\tau} - r_{f,\tau}, \qquad \widetilde{r}_{m,\tau} = r_{m,\tau} - r_{f,\tau}.$$

[1] We use τ for days and t for months. Use CRSP's daily value-weighted total market return and Ken French's daily RF for historical and realized betas.

The standard historical market beta estimator is

$$\widehat{\beta}_{i,t} = \frac{\widehat{\mathrm{Cov}}\left(\widetilde{r}_{i,\tau}, \widetilde{r}_{m,\tau}\right)}{\widehat{\mathrm{Var}}\left(\widetilde{r}_{m,\tau}\right)},$$

where the covariance and variance are calculated using observations τ in a specified historical estimation window.[2] Typically, the beta of a stock is estimated using trailing daily returns over some window: the past one year, three years, or the full available sample.

However, a beta estimated from historical returns is useful for portfolio construction only to the extent that it provides information about a stock's future market exposure. In this question, you will study how well beta estimates constructed using information available at time t predict the beta realized subsequently. Run the predictive regression

$$\widehat{\beta}^{\text{realized}}_{i,t+1} = \gamma_0 + \gamma_1 \widehat{\beta}^{\text{expected}}_{i,t} + u_{i,t+1},$$

where $\widehat{\beta}^{\text{realized}}_{i,t+1}$ is the beta realized over the following month and $\widehat{\beta}^{\text{expected}}_{i,t}$ is your estimate of the stock's beta using information available at time t. Construct $\widehat{\beta}^{\text{expected}}_{i,t}$ using the trailing-window estimators described above, as well as the estimator proposed by Ivo Welch in *Simply Better Market Betas*. In particular, implement Welch's slope-winsorized estimator with age weighting as an additional benchmark.[3]

For each beta estimator, report and compare the estimates of γ_0 and γ_1, the predictive-regression R^2, and the forecast error. A natural measure of prediction error is the weighted RMSE:

$$\text{RMSE} = \sqrt{\sum_{i \in \mathcal{S}_t} \omega_{i,t} \left(\widehat{\beta}^{\text{realized}}_{i,t+1} - \widehat{\beta}^{\text{expected}}_{i,t}\right)^2}.$$

where monthly weights $\omega_{i,t}$ are proportional to month-end t market capitalization, and $\mathcal{S}_t$ is the common stock sample across methods in month t. Compare that to the equal weighted RMSE.[4]

Interpret the results. How strongly do historical beta estimates predict subsequent realized betas? How does predictive performance vary across estimation windows and methods? What might explain differences in γ_1, R^2, and RMSE across the different beta estimators?[5]

Next, try to improve on the ability of Welch's method to predict each stock's beta in the following month. Propose an alternative method for forming expected betas and evaluate

[2] This estimator is numerically identical to the slope coefficient from the OLS market-model regression

$$\widetilde{r}_{i,\tau} = \alpha_i + \beta_i \widetilde{r}_{m,\tau} + \varepsilon_{i,\tau}.$$

[3] Ivo Welch (2022), "Simply Better Market Betas," *Critical Finance Review* 11(1), 37–64.

[4] To aggregate over time, average monthly MSEs first, then take the square root:

$$\text{RMSE}_{\text{overall}} = \sqrt{\frac{1}{T}\sum_{t}\sum_{i \in \mathcal{S}_t} \omega_{i,t} \left(\widehat{\beta}^{\text{realized}}_{i,t+1} - \widehat{\beta}^{\text{expected}}_{i,t}\right)^2}.$$

Here T counts evaluation months

[5] It may also be useful to sort stocks into portfolios based on historical beta and plot the average subsequent realized beta for each portfolio. This makes the persistence and mean reversion in beta considerably easier to see.

whether it improves out-of-sample predictive performance. Your proposed method should be motivated by an economic or statistical explanation of beta dynamics. In particular, think about what drives the relationship between historical beta estimates and next month's realized beta, and why a firm's exposure to the market may change over time.

Choose and justify a date that divides the sample into an in-sample period and a later out-of-sample period. Use only the earlier period to develop and fine-tune your method. Then evaluate both your method and Welch's method separately in the in-sample and out-of-sample periods. Report the prediction errors for each method, together with uncertainty in the difference in their performance. Assess the mean monthly paired difference in weighted squared loss (your method minus Welch's) on the same stocks, using Newey–West standard errors.

Document the approaches you tried and explain why you expected them to work. Use the empirical results to evaluate that reasoning, including cases in which an apparent in-sample improvement fails to persist out of sample.

Question 2 — Forecasting Factor and Stock Returns

Forecasting a stock's factor exposures is only one part of forecasting its return. We also need forecasts of the factor returns themselves. A benchmark is to use the historical average realized return of each factor. In this question, we ask whether you can improve on that benchmark for the market, value, and size premiums, and whether better factor-return forecasts translate into better forecasts of individual stock returns.

Use the traded Mkt–RF, HML, and SMB factors from Homework 1 (or from Ken French's website). For each stock, write the three-factor model as

$$r^e_{i,t+1} = \beta_{i,M,t+1}\,\text{Mkt–RF}_{t+1} + \beta_{i,HML,t+1}\,\text{HML}_{t+1} + \beta_{i,SMB,t+1}\,\text{SMB}_{t+1} + \varepsilon_{i,t+1}.$$

Your forecast of next month's excess return is

$$\widehat{r}^e_{i,t+1|t} = \widehat{\boldsymbol{\beta}}'_{i,t+1|t}\widehat{\boldsymbol{\lambda}}_{t+1|t},$$

where $\widehat{\boldsymbol{\beta}}_{i,t+1|t}$ contains the stock's forecast exposures to Mkt–RF, HML, and SMB, and $\widehat{\boldsymbol{\lambda}}_{t+1|t}$ contains the corresponding forecasts of the three factor returns.

As a benchmark, estimate the factor premia using historical averages, and then see whether alternative forecasts of the three factor returns improve upon these historical-average forecasts and, ultimately, whether they reduce the prediction error for individual stock returns.

Explain how you estimate the factor exposures and factor-return forecasts using only information available at the forecast date.[6] Begin by forecasting each stock's excess return using the long term average excess return on the CRSP Value Weighted Return Index (which is typically much better than using the own stock historical average), and compare this with the factor-model forecast when each factor premium is estimated by its expanding historical average. All averages must use only observations available at the time. How well do these simple forecasts predict the following month's realized stock returns?

[6] Note that these are joint exposures to the three traded factors. A market beta estimated from the single-factor regression in Question 1 need not equal the market loading from the three-factor regression. Why?

Next, develop an approach to forecasting the three factor premiums that you expect to improve on their historical averages. Explain the economic reason for your approach. For example, differences in valuations between value and growth stocks might contain information about the expected value premium; the literature on expected returns offers several ways to think about such relationships.[7]

Think of these as starting points. You are not expected to use any particular forecasting model. As in Question 1, develop and fine-tune your method using an initial in-sample period, then evaluate it in a later out-of-sample period. Compare mean squared prediction errors for each factor forecast and for the resulting stock-return forecasts, separately in the in-sample and out-of-sample periods. A useful summary measure is the out-of-sample pseudo-R^2 of Goyal and Welch (2008),

$$R^2_{\text{OOS}} = 1 - \frac{\text{MSE}_{\text{model}}}{\text{MSE}_{\text{benchmark}}},$$

where a positive value indicates that the model improves on the benchmark and a negative value indicates that it performs worse.[8]

Hold the beta forecasts fixed across the methods so that this comparison isolates the contribution of forecasting factor returns. Use the same observations and weighting for each comparison. Discuss the magnitude and statistical uncertainty of any improvement. What do your results tell you about the difficulty of estimating expected returns from noisy realized returns?

Question 3 — Constructing Pure Factors

A stock that looks cheap may also be small, unprofitable, or particularly sensitive to market returns. A successful value strategy may therefore earn returns from many sources. This question asks how to differentiate the return associated with a characteristic from the returns associated with other characteristics that tend to accompany it.

Using your Homework 1 data, estimate the monthly cross-sectional regression

$$r^e_{t+1} = X_t f_{t+1} + u_{t+1}$$

Here r^e_{t+1} contains stocks' next-month excess returns and X_t contains industry indicators (with no separate intercept), and standardized log market capitalization, log book-to-market, profitability, investment, and the predicted market beta from Question 1. Use Fama–French 12 industries throughout the sample, applying the mapping in Ken French's Data Library to historical CRSP SIC codes at each month-end.[9] Estimate each monthly regression by weighted least squares, using the square root of market capitalization at the end of month t as the regression weight.[10]

[7] See Randolph B. Cohen, Christopher Polk, and Tuomo Vuolteenaho (2003), "The Value Spread," *Journal of Finance* 58(2), 609–641; and John Y. Campbell and Samuel B. Thompson (2008), "Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average?" *Review of Financial Studies* 21(4), 1509–1531.

[8] Amit Goyal and Ivo Welch (2008), "A Comprehensive Look at the Empirical Performance of Equity Premium Prediction," *Review of Financial Studies* 21(4), 1455–1508. As in Question 1, your MSE can be equal weighted or value-weighted.

[9] French's published industry portfolios assign industries annually.

[10] Here the weight is the coefficient on the squared residual in the WLS objective: $\sum_i \sqrt{\text{ME}_{i,t}}\, u^2_{i,t+1}$. Use the same weighting when assessing orthogonality below.

The estimated coefficients $\widehat{f}_{t+1}$ are the factor returns in this model: each measures the month's return associated with one unit of a characteristic, holding the other included characteristics fixed. What does the average value coefficient tell you, for instance, about cheap stocks relative to otherwise similar stocks? Explain how this interpretation depends on the units of the value measure and the controls included in X_t. Re-express the characteristics using a different normalization that preserves their information content, and compare the coefficients, fitted returns, and residuals. What is the difference between a change in measurement units versus a change in the economic comparison, such as omitting profitability from the regression?

Next, choose a signal z_t that you believe predicts returns, say a value signal such as the Book-to-Market ratio, or earnings-to-price. Identify the exposures you want to remove, and collect those columns of X_t in a control matrix C_t. If your controls do not include the full set of industry indicators, add an intercept to C_t. Explain why removing them would make the signal more informative about its own return-predictive content.

At each month-end, regress z_t on *all* the columns of C_t jointly, using the same WLS weights. Retain the residual as your orthogonalized signal:

$$z_t^{\perp} = z_t - C_t \widehat{a}_t, \qquad C_t' W_t z_t^{\perp} = 0, \qquad W_t = \mathrm{diag}(w_{i,t}),$$

where $\widehat{a}_t$ contains the fitted coefficients. Verify the orthogonality condition ($C_t' W_t z_t^{\perp} = 0$) numerically. A positive residual identifies a stock whose signal is higher than its controls would predict. Report how much of the signal's weighted cross-sectional variation the controls explain. Here "pure" means zero weighted linear exposure to the selected controls.

Finally, using characteristics observed at the end of t and returns realized in $t+1$, estimate the two cross-sectional regressions

$$r_{t+1}^e = C_t \delta_{t+1} + \gamma_{t+1} z_t + u_{t+1},$$
$$r_{t+1}^e = \alpha_{t+1}^{\perp} \mathbf{1} + \gamma_{t+1}^{\perp} z_t^{\perp} + v_{t+1}.$$

The first regression includes the original signal and the controls C_t. The second includes only the pure signal and an intercept. Use exactly the same stocks and WLS weights in both regressions and in the projection used to construct $z_t^{\perp}$.[11]

Compare the coefficient on z_t in the first regression with the coefficient on $z_t^{\perp}$ in the second. You'll find: $\widehat{\gamma}_{t+1} = \widehat{\gamma}_{t+1}^{\perp}$ in each month. You have changed the signal and removed the controls from the return regression, yet the coefficient is unchanged. Is this an empirical finding about your signal, or does it hold by construction?

Now think about what purification changes about the economic question. For example, a valuation signal may predict returns because it identifies cheap stocks, because cheap stocks tend to be small or concentrated in particular industries, or because both matter. What does the original signal tell you about the returns to buying stocks with that characteristic, and what does the pure signal tell you about differences in returns among stocks with similar controls? What can this tell us about the source of the return premium?

[11] Keep the pure signal in the units of the projection residual; do not standardize it again.